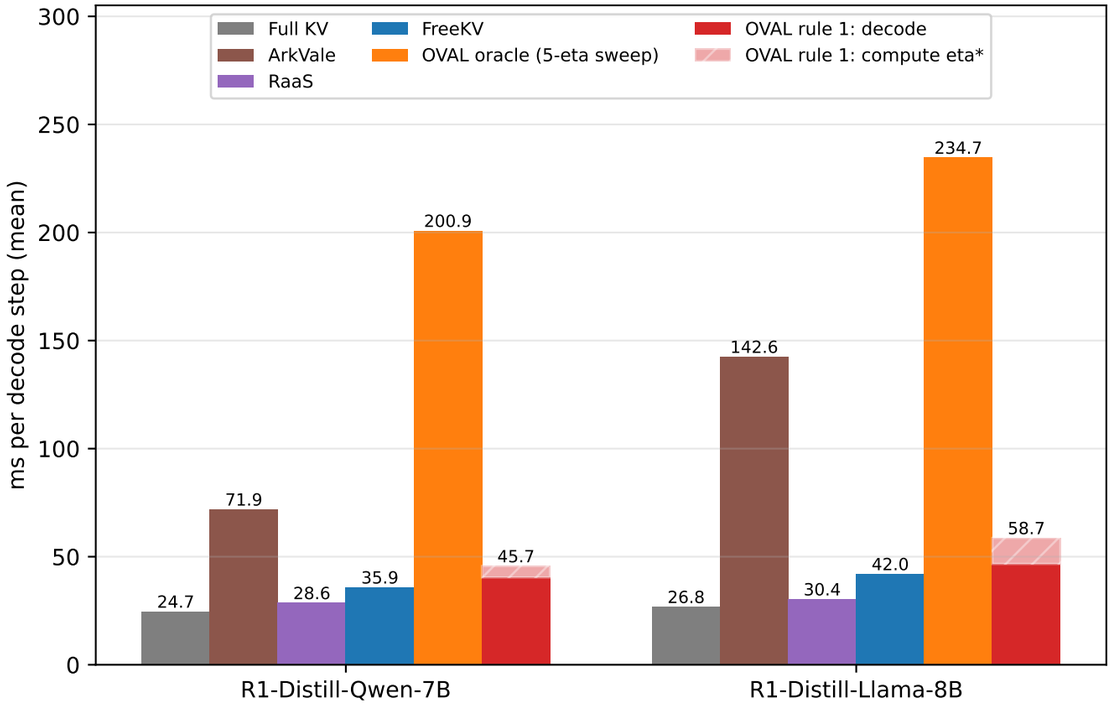
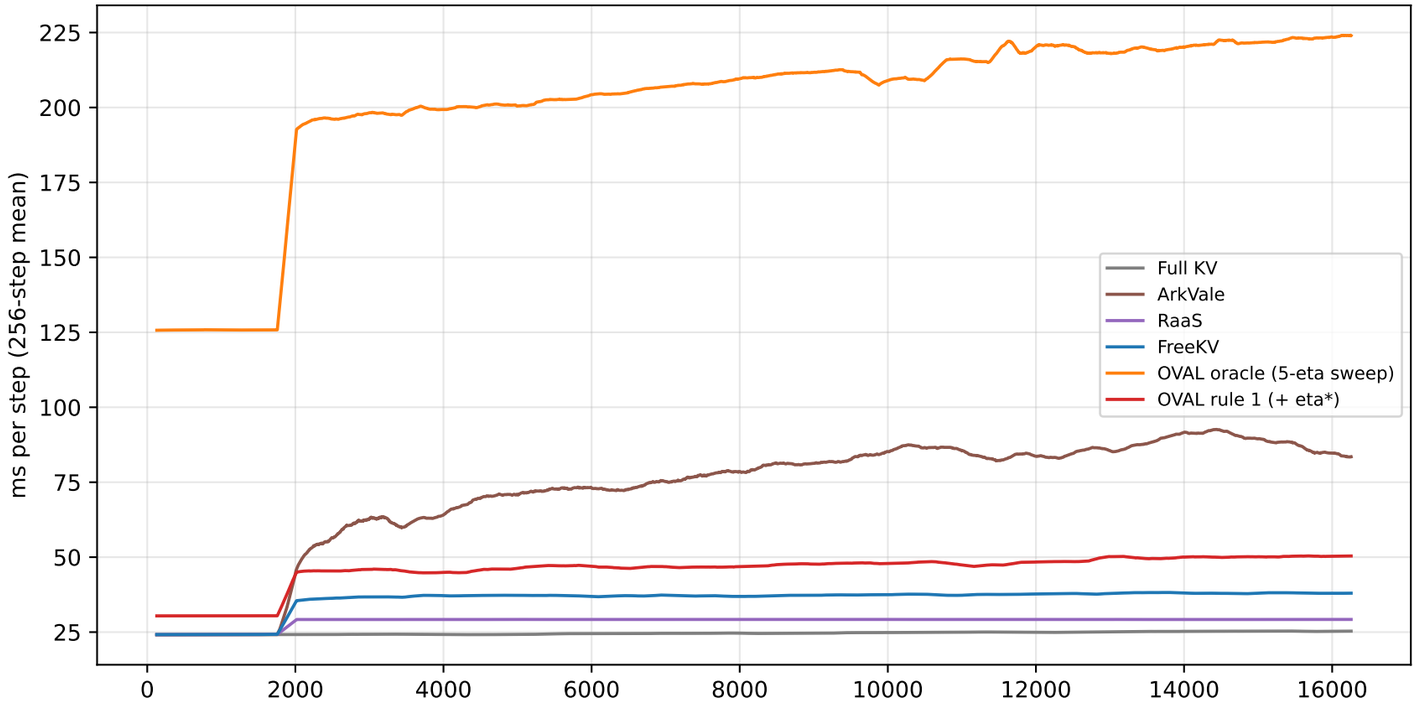
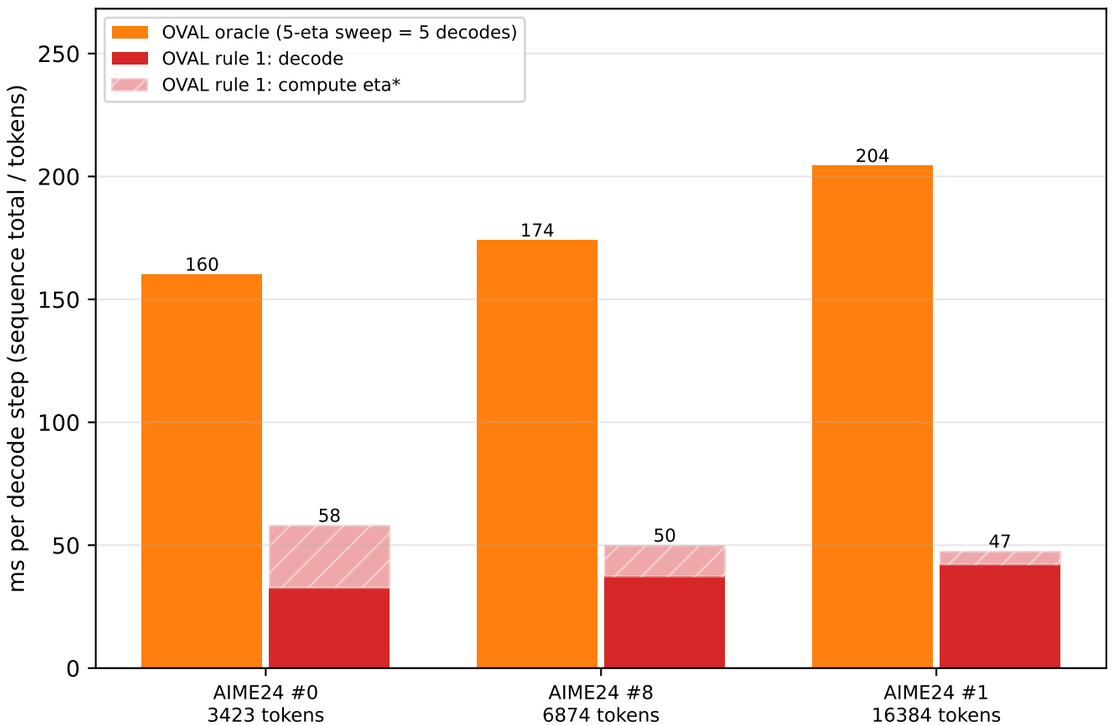
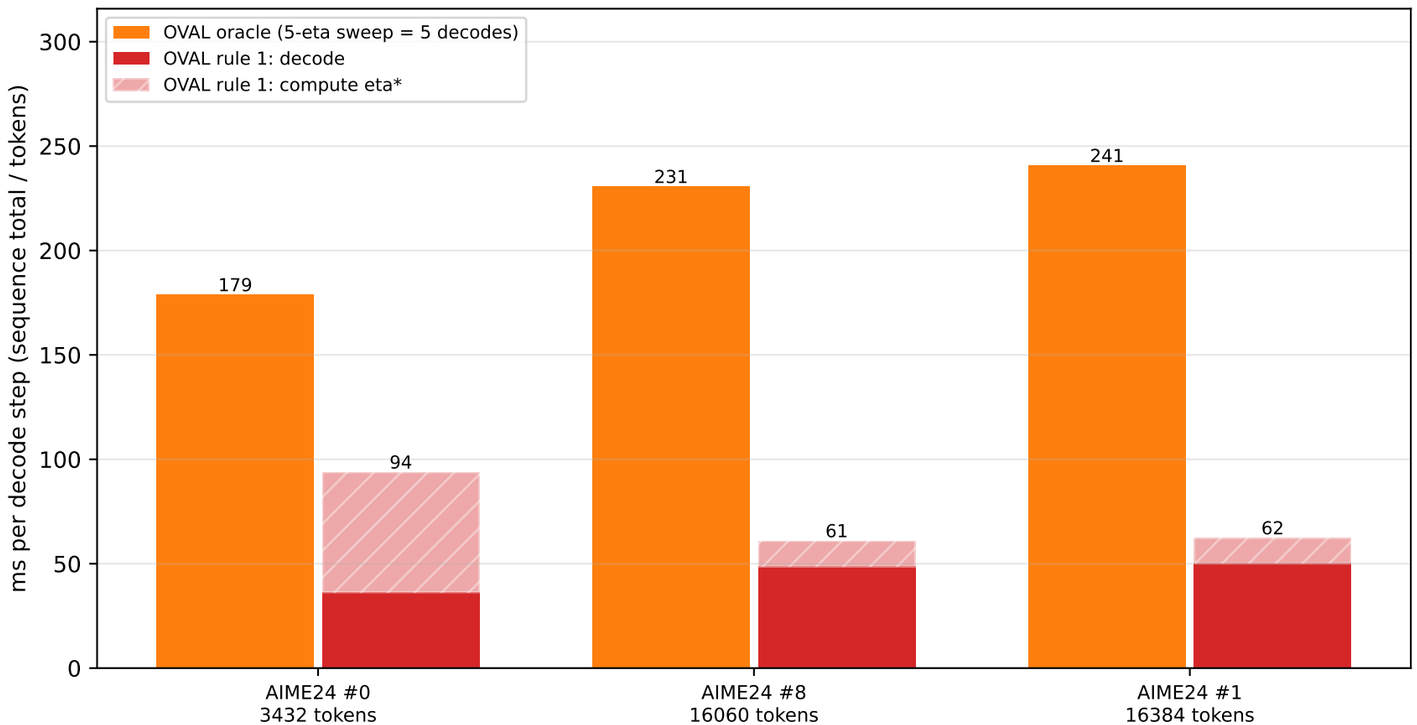

Decode latency, AIME24 problem 0, 16384 decoded tokens, batch 1
budget 2048 tokens (incl. 512 sink + 512 recent), A6000
Full KV
ArkVale
RaaS
FreeKV
OVAL oracle (5-eta sweep)
OVAL rule 1: decode
OVAL rule 1: compute eta*
ms per decode step (mean)
24.7
71.9
28.6
35.9
200.9
45.7
26.8
142.6
30.4
42.0
234.7
58.7
R1-Distill-Qwen-7B
R1-Distill-Llama-8B


R1-Distill-Qwen-7B: per-step decode time over the generation
rule 1 includes its eta* time spread evenly over the steps
ms per step (256-step mean)
decode step
Full KV
ArkVale
RaaS
FreeKV
OVAL oracle (5-eta sweep)
OVAL rule 1 (+ eta*)


R1-Distill-Qwen-7B: OVAL oracle (sweep 5 etas) vs OVAL rule 1
OVAL oracle (5-eta sweep = 5 decodes)
OVAL rule 1: decode
OVAL rule 1: compute eta*
ms per decode step (sequence total / tokens)
160
58
174
50
204
47
AIME24 #0
3423 tokens
AIME24 #8
6874 tokens
AIME24 #1
16384 tokens


R1-Distill-Llama-8B: OVAL oracle (sweep 5 etas) vs OVAL rule 1
OVAL oracle (5-eta sweep = 5 decodes)
OVAL rule 1: decode
OVAL rule 1: compute eta*
ms per decode step (sequence total / tokens)
179
94
231
61
241
62
AIME24 #0
3432 tokens
AIME24 #8
16060 tokens
AIME24 #1
16384 tokens